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Mobile software automation method based on semantic knowledge graph and LLM enhanced reasoning (English)
基于语义知识图谱与LLM增强推理的移动软件自动化方法 (Chinese)

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Abstract (Chinese)

本发明提出一种基于语义知识图谱与LLM增强推理的移动软件自动化方法，通过自动化界面探索构建初始知识图谱，并利用大语言模型的强语义理解能力对图谱进行高效补全与语义富集，快速形成涵盖组件属性、操作逻辑及跨应用依赖的完整知识表示；依托知识图谱的结构化知识对大语言模型的推理过程进行有效约束，系统通过混合遍历策略自动探索应用界面并构建知识图谱，采用模板-实例抽象机制适配动态内容，并基于图谱进行多维度指令匹配与路径规划，最终将自然语言任务转化为可执行的系统级操作序列。本发明实现了从指令解析到执行监控的端到端自动化闭环，能够覆盖多样化的用户操作场景，在保障响应效率的同时，大幅降低了解析与执行误差，有效提升了移动端自动化控制的适应性、鲁棒性与用户体验。

[Image Omitted]

Abstract (English)

The invention provides a mobile software automation method based on semantic knowledge graph and LLM enhanced reasoning, which comprises the steps of constructing an initial knowledge graph through an automation interface exploration, carrying out high-efficiency complementation and semantic enrichment on the graph by utilizing the strong semantic understanding capability of a large language model, quickly forming complete knowledge representation covering component attributes, operation logics and cross application dependence, carrying out effective constraint on the reasoning process of the large language model by relying on the structured knowledge of the knowledge graph, automatically exploring an application interface through a mixed traversal strategy by a system, constructing the knowledge graph, adapting dynamic content by adopting a template-instance abstract mechanism, carrying out multidimensional instruction matching and path planning based on the graph, and finally converting a natural language task into an executable system-level operation sequence. The invention realizes end-to-end automatic closed loop from instruction analysis to execution monitoring, can cover diversified user operation scenes, greatly reduces analysis and execution errors while guaranteeing response efficiency, and effectively improves adaptability, robustness and user experience of automatic control of the mobile terminal.

[Image Omitted]

Synopsis AI Document Summarization (Default)

Summary Size: Default

Key Phrases

- semantic knowledge graph
- mobile software automation
- language instruction
- software automation method based

Mobile software automation method based on semantic knowledge graph and LLM enhanced reasoning

Technical Field

The present invention relates to the field of artificial intelligence, software automation (Software Automation) and mobile computing technologies, and in particular, to a mobile software automation method, apparatus, electronic device, computer readable storage medium and computer program product based on semantic knowledge graph and LLM enhanced reasoning.

More particularly, the invention relates to a mobile application Graphical User Interface (GUI) automation technology including automation testing, task script execution and exploratory interactions, knowledge graph construction and application technology, knowledge modeling methods used in software engineering to represent user interface structures, component semantics and interaction logic, reasoning and decision applications in software automation involving Large Language Models (LLMs) including how to understand interface semantics, parse user intent and generate reliable operation sequences using LLMs, and cross-application, multi-step mobile end task automation technology aimed at solving end-to-end operation execution problems under conditions of interface dynamic changes, lack of stable API interfaces. The present invention aims to provide an intelligent autonomous automation solution for mobile applications that is capable of adapting to interface changes without relying on source code, requiring no predefined scripts or fixed APIs.

The invention relates to a mobile software automation method based on structural semantic knowledge graph and LLM enhanced reasoning, belonging to the technical field of artificial intelligence and mobile computing intersection. The method aims to solve the problems of poor robustness, insufficient maintainability, inaccurate analysis of natural language instructions and the like caused by frequent interface change, uncontrollable dynamic content and high script maintenance cost in the current mobile application automation technology. The method is characterized by comprising the steps of realizing deep synergy of a large language model and a semantic knowledge graph, on one hand, constructing an initial knowledge graph through an automatic interface exploration, carrying out high-efficiency complementation and semantic enrichment on the graph by utilizing the strong semantic understanding capability of the large language model, and rapidly forming complete knowledge representation covering component attributes, operation logics and cross-application dependencies, and on the other hand, carrying out effective constraint on the reasoning process of the large language model by relying on the structured knowledge of the knowledge graph, thereby inhibiting illusion generation and remarkably improving the accuracy and reliability of instruction analysis. Wherein the illusion refers to, for example, when the knowledge graph stores an explicit entity such as "mobile application page element", a relationship such as "button a→trigger function B", and an attribute such as "format requirement of input box C is a cell phone number". If the model generates the content such as a function B associated button D and an input box C supporting a mailbox format which are not existed in the map during reasoning, the output of the model has illusion. The system automatically explores an application interface through a mixed traversal strategy, constructs a knowledge graph, adapts dynamic content by adopting a template-instance abstraction mechanism, performs multidimensional instruction matching and path planning based on the graph, and finally converts a natural language task into an executable system-level operation sequence. The method realizes end-to-end automatic closed loop from instruction analysis to execution monitoring, can cover diversified user operation scenes, greatly reduces analysis and execution errors while guaranteeing response efficiency, and effectively improves adaptability, robustness and user experience of automatic control of the mobile terminal.

Aiming at the defects of the prior art, as shown in fig. 6, the invention provides a mobile software automation method based on semantic knowledge graph and LLM enhanced reasoning, which comprises the following steps:

An initial knowledge graph construction step, namely extracting interface information when mobile equipment runs mobile Application (APP), and exploring and obtaining a unique page signature of each access page according to the content and the function of each access page in the interface information to construct an initial knowledge graph [Image Omitted] The node set V comprises page nodes and interaction component nodes thereof, and the edge set E comprises structural edges reflecting father-son inclusion relation and action edges for recording operation jumps;

The mobile software automation method based on semantic knowledge graph and LLM enhanced reasoning comprises the following steps:

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normalizing text attribute of component, comprising [Image Omitted] Generating semantically equivalent alias sets for supporting subsequent fuzzy matching, combining structural features of the pages and jump histories, dividing the pages into three types of normal, temporary and cyclic through formulas, [Image Omitted] The logical association and the cross-page function dependence among components which are not directly observed in the inferred exploration process are written into the graph database, and finally the enhanced knowledge graph is formed [Image Omitted] .

The mobile software automation method based on semantic knowledge graph and LLM enhanced reasoning comprises the following steps:

traversing the complete knowledge graph [Image Omitted] If the nodes exceeding the preset value have the same father node, layout structure and role type, only the text and image attributes have difference, and the area supports the scrolling operation, judging the dynamic area [Image Omitted] In the automatic execution process of the execution and monitoring steps, the dynamic instance in the current page is extracted in real time through a system level command [Image Omitted] Instances are matched by structural similarity [Image Omitted] Binding with the corresponding template node T, and calculating the matching score of the instance and the instruction through a formula when the user instruction relates to the dynamic target:

The mobile software automation method based on semantic knowledge graph and LLM enhanced reasoning comprises the following steps:

converting instruction operation sequences into system-level input events, extracting a control tree of a current page through a system-level command after each operation is completed in the execution process, and calculating page signatures [Image Omitted] Signature with expected page [Image Omitted] Comparing, if it meets [Image Omitted] The $\text{sim}()$ represents the page signature similarity function defined before, and the θ represents the state matching threshold value, so that the current operation is judged to be successfully executed, the next action is continuously executed, and otherwise, an exception handling mechanism is triggered;

the instruction analysis and graph constraint reasoning module receives natural language instructions input by a user [Image Omitted] Analyzing, performing multidimensional semantic grounding, and based on a complete knowledge graph [Image Omitted] Calculating natural language instructions by integrating multiple dimensions such as text similarity, semantic embedding correlation, component role compatibility and interface context [Image Omitted] Matching score with candidate UI elements [Image Omitted] Selecting candidate UI elements with highest scores as target components, performing graph-based action planning, and modeling a task completion process as in [Image Omitted] The path searching problem searches the shortest path formed by legal action edges by taking the current state node as a starting point and the grounding target node as an end point, and generates an instruction operation sequence;

The functions, if implemented in the form of software functional units and sold or used as a stand-alone product, may be stored in a computer-readable storage medium. Based on this understanding, the technical solution of the present invention may be embodied essentially or in a part contributing to the prior art or in a part of the technical solution, in the form of a software product stored in a storage medium, comprising several instructions for causing a computer device (which may be a personal computer, a server, a network device, etc.) to perform all or part of the steps of the method according to the embodiments of the present invention. The storage medium includes a U disk, a removable hard disk, a read-only memory (ROM), a random access memory (random access memory, RAM), a magnetic disk, an optical disk, or other various media capable of storing program codes.

The invention provides a mobile software autonomous automation method based on structured semantic knowledge graph and LLM enhanced reasoning, which is characterized in that a collaborative framework of automatic UI exploration and structured modeling, semantic knowledge graph enhancement, constraint reasoning and planning-robust execution monitoring is adopted, so that the technical pain points of poor interface variability adaptability, inaccurate analysis of natural language instructions, difficult dynamic content processing, difficult multiplexing of automated assets and the like in the existing mobile automation technology are solved, and the mobile terminal autonomous automation with high precision, high robustness and maintainability is realized.

the semantic enhancement and knowledge enrichment module is used for carrying out initial knowledge graph according to a lightweight large language model LLM and a scene prompt template [Image Omitted] Performing deep semantic completion, deducing functional semantic roles of the nodes according to the attributes of the components by using LLM, normalizing the text of the nodes to generate a

group of equivalent semantics for supporting fuzzy matching, combining the initial knowledge graph [Image Omitted] Classifying interface types by LLM to obtain the interface types belonging to normal, temporary or cyclic categories, combining interface types, equivalent semantics and functional semantics roles, and utilizing LLM to infer logical association among components and cross-page function dependence which are not directly observed in exploration so as to complement the initial knowledge graph [Image Omitted] Obtaining a semantically enhanced complete knowledge graph [Image Omitted] ;

Dynamic content processing module, in complete knowledge graph [Image Omitted] Introducing a representation mechanism of template instance separation, and identifying a complete knowledge graph [Image Omitted] As a structural template node [Image Omitted] Recording its fixed layout mode, component role and semantic label, and using specific dynamic item as run-time example [Image Omitted] Contains the text and image characteristics which change in real time and is connected with the corresponding template nodes through a dynamic binding mechanism [Image Omitted] Associating;

the instruction analysis and graph constraint reasoning module receives natural language instructions input by a user [Image Omitted] Analyzing, performing multidimensional semantic grounding, and based on a complete knowledge graph [Image Omitted] Calculating natural language instructions by integrating multiple dimensions such as text similarity, semantic embedding correlation, component role compatibility and interface context [Image Omitted] Matching score with candidate UI elements [Image Omitted] Selecting candidate UI elements with highest scores as target components, performing graph-based action planning, and modeling a task completion process as in [Image Omitted] The path searching problem searches the shortest path formed by legal action edges by taking the current state node as a starting point and the grounding target node as an end point, and generates an instruction operation sequence;

Document Description

Mobile software automation method based on semantic knowledge graph and LLM enhanced reasoning

Technical Field

The present invention relates to the field of artificial intelligence, software automation (Software Automation) and mobile computing technologies, and in particular, to a mobile software automation method, apparatus, electronic device, computer readable storage medium and computer program product based on semantic knowledge graph and LLM enhanced reasoning.

More particularly, the invention relates to a mobile application Graphical User Interface (GUI) automation technology including automation testing, task script execution and exploratory interactions, knowledge graph construction and application technology, knowledge modeling methods used in software engineering to represent user interface structures, component semantics and interaction logic, reasoning and decision applications in software automation involving Large Language Models (LLMs) including how to understand interface semantics, parse user intent and generate reliable operation sequences using LLMs, and cross-application, multi-step mobile end task automation technology aimed at solving end-to-end operation execution problems under conditions of interface dynamic changes, lack of stable API interfaces. The present invention aims to provide an intelligent autonomous automation solution for mobile applications that is capable of adapting to interface changes without relying on source code, requiring no predefined scripts or fixed APIs.

Background

The software automation (Software Automation) is a computer application field for realizing the automation of program design and document design by a technical means, and the core aim is to improve the software development and management efficiency.

With the advanced development of the mobile internet, mobile application has become a core carrier for daily life and work of people. With this, there is a strong demand for efficient and reliable automation of mobile applications, which have application scenarios that widely cover software testing, barrier-free assistance, process automation, personal task assistants, etc. However, the inherent features of mobile applications, such as frequent interface updates, multiple screen sizes and resolutions, rich dynamic content, and operating system version fragmentation, present significant challenges to traditional automation techniques.

Existing mobile application automation technologies mainly follow the following technical paths, each with significant limitations. First, scripting and recording playback tools, represented by Appium, uiAutomator, etc., require developers or testers to write interactive scripts based on UI element locators, or to record the operational process generation scripts. The main problem is vulnerability, once the UI layout, component identifier or structure of the application is changed, the script is invalid, and the elements are needed to be manually repositioned or the script is modified, so that the maintenance cost is high. Furthermore, they have difficulty handling unexpected dialog boxes or dynamically loaded content that pops up at run-time. Secondly, modeling and random exploration methods are adopted, and the methods automatically or semi-automatically explore application interfaces through programs to construct state machines, activity diagrams or event stream models so as to improve the coverage rate of the interfaces. For example, monkey-based random testing, deep learning-based GUI traversal strategies, and the like. While they can discover more interface states, there is a general lack of advanced semantic understanding capabilities, whose optimization goal is typically to maximize the Activity or Widget coverage, etc., structural indicators, rather than reliably completing user-specified semantic-level tasks. They cannot understand the meaning of "commit" of buttons nor handle complex workflows that need to make context-dependent decisions across multiple interfaces. Third is an API and intent based Assistant, such as google asside, phone vendor's voice Assistant, that interacts through a predefined shortcut or deep link with the application. The method provides a natural language input channel, but the capability of the method is completely limited by the pre-registered and published API interfaces of a developer, any function or interface of an unexposed interface in the application cannot be operated, and the method is not truly automated at the interface level per se, so that the universality is seriously insufficient.

In recent years, large language models represented by ChatGPT exhibit powerful natural language understanding and contextual reasoning capabilities, have been introduced into the field of GUI automation. Some leading edge studies attempt to have the LLM analyze the screen shots or control trees, directly generate the next operation instructions, or generate the exploration scene in the test. These works demonstrate that LLM has the potential for preliminary interface element recognition and intent reasoning and can produce more humanoid interactions than random exploration. However, the current LLM-driven mobile automation scheme still has the key defects of lack of a persistent and structured interface model, high grounding failure rate when UI text labels are not completely consistent with the expression of the user instructions, poor adaptability to dynamic content and interface changes, concentration of test generation or static interface analysis, lack of systematic processing strategies for dynamic lists, personalized recommendations and layout adjustment brought by application version update in actual use, lack of sense and unreliable reasoning, lack of sense and unreliable processing strategies for

grounding user instructions to UI elements, lack of deep semantic labeling of component function roles, lack of a robust processing mechanism for synonyms and dynamic content, and lack of automatic constraint on LLM, failure of executing the current operation process or error operation interruption due to the fact that the UI text labels are not completely consistent with the expression of the user instructions, poor adaptability to dynamic content and interface changes, and most researches are concentrated on test generation or static interface analysis.

In view of the foregoing, the prior art has not yet provided an autonomous mobile terminal automation scheme with high robustness, high accuracy, strong adaptability and good maintainability. Therefore, a new method capable of fusing structured knowledge representation and semantic understanding reasoning is proposed to break through the bottleneck.

Disclosure of Invention

The invention relates to a mobile software automation method based on structural semantic knowledge graph and LLM enhanced reasoning, belonging to the technical field of artificial intelligence and mobile computing intersection. The method aims to solve the problems of poor robustness, insufficient maintainability, inaccurate analysis of natural language instructions and the like caused by frequent interface change, uncontrollable dynamic content and high script maintenance cost in the current mobile application automation technology. The method is characterized by comprising the steps of realizing deep synergy of a large language model and a semantic knowledge graph, on one hand, constructing an initial knowledge graph through an automatic interface exploration, carrying out high-efficiency complementation and semantic enrichment on the graph by utilizing the strong semantic understanding capability of the large language model, and rapidly forming complete knowledge representation covering component attributes, operation logics and cross-application dependencies, and on the other hand, carrying out effective constraint on the reasoning process of the large language model by relying on the structured knowledge of the knowledge graph, thereby inhibiting illusion generation and remarkably improving the accuracy and reliability of instruction analysis. Wherein the illusion refers to, for example, when the knowledge graph stores an explicit entity such as "mobile application page element", a relationship such as "button a→trigger function B", and an attribute such as "format requirement of input box C is a cell phone number". If the model generates the content such as a function B associated button D and an input box C supporting a mailbox format which are not existed in the map during reasoning, the output of the model has illusion. The system automatically explores an application interface through a mixed traversal strategy, constructs a knowledge graph, adapts dynamic content by adopting a template-instance abstraction mechanism, performs multidimensional instruction matching and path planning based on the graph, and finally converts a natural language task into an executable system-level operation sequence. The method realizes end-to-end automatic closed loop from instruction analysis to execution monitoring, can cover diversified user operation scenes, greatly reduces analysis and execution errors while guaranteeing response efficiency, and effectively improves adaptability, robustness and user experience of automatic control of the mobile terminal.

Aiming at the defects of the prior art, as shown in fig. 6, the invention provides a mobile software automation method based on semantic knowledge graph and LLM enhanced reasoning, which comprises the following steps:

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Semantic enhancement and knowledge enrichment, namely, according to a lightweight large language model LLM and a scene prompt template, performing initial knowledge graph [Image Omitted] Performing deep semantic completion, deducing functional semantic roles of the nodes according to the attributes of the components by using LLM, normalizing the text of the nodes to generate a group of equivalent semantics for supporting fuzzy matching, combining the initial knowledge graph [Image Omitted] Classifying interface types by LLM to obtain the interface types belonging to normal, temporary or cyclic categories, combining interface types, equivalent semantics and functional semantics roles, and utilizing LLM to infer logical association among components and cross-page function dependence which are not directly observed in exploration so as to complement the initial knowledge graph [Image Omitted] Obtaining a semantically enhanced complete knowledge graph [Image Omitted] ;

Dynamic content processing step, in the complete knowledge graph [Image Omitted] Introducing a representation mechanism of template instance separation, and identifying a complete knowledge graph [Image Omitted] As a structural template node [Image Omitted] Recording its fixed layout mode, component role and semantic label, and using specific dynamic item as run-time example [Image Omitted] Contains the text and image characteristics which change in real time and is connected with the corresponding template nodes through a dynamic binding mechanism [Image Omitted] Associating;

instruction analysis and graph constraint reasoning step, receiving natural language instruction input by user [Image Omitted] Analyzing, performing multidimensional semantic grounding, and based on a complete knowledge graph [Image Omitted] Calculating natural language instructions by integrating multiple dimensions such as text similarity, semantic embedding correlation, component role compatibility and interface context [Image Omitted] Matching score with candidate UI elements [Image Omitted] Selecting candidate UI elements with highest scores as target components, performing graph-based action planning, and modeling a task completion process as in [Image Omitted] The path searching problem searches the shortest path formed by legal action edges by taking the current state node as a starting point and the grounding target node as an end point, and generates an instruction operation sequence;

Executing and monitoring, namely converting the abstract action sequence of the instruction operation sequence into a system-level input event comprising multi-step operation so as to automatically execute the abstract action sequence on the mobile device, and monitoring the change of interface information in the executing process, re-acquiring a current control tree, calculating page signatures and comparing the current control tree with an expected state after each step of operation [Image Omitted] 。

The mobile software automation method based on semantic knowledge graph and LLM enhanced reasoning comprises the following steps:

a knowledge graph increment maintenance step, in which newly discovered interface states, components and interaction relations are continuously fed back and updated to complete knowledge graphs in the automatic execution process [Image Omitted] The online learning and incremental expansion of the atlas are realized, so that the system has the capability of continuous evolution.

The mobile software automation method based on semantic knowledge graph and LLM enhanced reasoning comprises the steps of exploring the content and the function of each accessed page in interface information to ensure the initial knowledge graph [Image Omitted] Pruning the interface information, including integrating the contract-level information through text merging, transmitting dynamic state through attribute inheritance mechanism, and eliminating invalid nodes with empty text, messy codes and abnormal coordinates.

The mobile software automation method based on semantic knowledge graph and LLM enhanced reasoning comprises the following steps:

Initial knowledge graph [Image Omitted] Center node set [Image Omitted] Edge set [Image Omitted] , [Image Omitted] For a structural edge, i.e. which nodes are contained in the page, [Image Omitted] Is an action edge, and when the APP program runs, an application initial starting page is started [Image Omitted] Inclusion exploration queue [Image Omitted] I.e. [Image Omitted] Initializing a set of accessed page signatures [Image Omitted] The action edge types comprise clicking, text input, sliding and returning operations;

Queue pair [Image Omitted] Each page of (3) [Image Omitted] The control tree is extracted by a system level command [Image Omitted] Calculating the unique page signature based on the layout structure, the text content and the component type as:

[Image Omitted]

wherein, the [Image Omitted] Representing pages [Image Omitted] The unique structure signature of the (a) is used for page duplication removal and identity recognition; [Image Omitted] representing pages [Image Omitted] The text token set of (2) is generated by normalizing visible component text; [Image Omitted] representing pages [Image Omitted] By mapping control trees to layout hash values of (a) [Image Omitted] Performing hash calculation to obtain the data for capturing the parent-child hierarchical relationship and ordering characteristics of the components; [Image Omitted] representing pages [Image Omitted] The component role sets are extracted from the component attributes and are determined after LLM judgment and used for representing the component function types;

Calculating the similarity of the page signatures through a weighted combination formula, and judging whether the page has been accessed or not:

[Image Omitted]

wherein, the [Image Omitted] Representing pages [Image Omitted] Is signed by (a) [Image Omitted] And page [Image Omitted] Is signed by (a) [Image Omitted] The similarity between the two is in the value range of 0, 1; [Image Omitted] the weight coefficients of the text similarity, the layout consistency and the role similarity respectively satisfy the following conditions [Image Omitted] ; [Image Omitted] Representing a Jacquard similarity function for calculating the degree of overlap of the two sets; [Image Omitted] Respectively are pages [Image Omitted] A text token set of (a); [Image Omitted] Respectively are pages [Image Omitted] $1[-J]$ is an indication function, returns 1 when the condition in brackets is satisfied, otherwise returns 0; [Image Omitted] Respectively are pages [Image Omitted] If the similarity reaches a preset threshold, determining to repeat the page, skipping the subsequent processing, otherwise, adding the page signature into the accessed set [Image Omitted] ;

Enumerating an operational action set for a current page [Image Omitted] Executing actions one by one [Image Omitted] Observing the new page generated [Image Omitted] Face, identify the temporary pages such as popup window, permission prompt, etc. through heuristic rules, if [Image Omitted] Fewer number of operable components, cancellation, permission of feature text, etc., and return to the original page after performing the return action [Image Omitted] Judging that the page is a temporary page, not recording the jump relation, otherwise recording the jump relation [Image Omitted]) And will [Image Omitted] Enqueue [Image Omitted] Executing a system return action back to the page [Image Omitted] ;

Constructing an initial knowledge graph based on the exploration result [Image Omitted] For all the nodes corresponding to the unique pages, the page signature is taken as the unique identification, [Image Omitted] For visible and interactable component nodes in the control tree, each component node satisfies:

[Image Omitted]

Wherein the method comprises the steps of [Image Omitted] The text is displayed for the component and, [Image Omitted] Is a unique identification for the component and, [Image Omitted] For a range of coordinates of the component, [Image Omitted] The interaction attributes of the components are respectively, [Image Omitted] Is a component type attribute.

The mobile software automation method based on semantic knowledge graph and LLM enhancement reasoning comprises the following steps:

a lightweight large language model and a scene prompt template are introduced, and an initial knowledge graph is mapped [Image Omitted] Semantic completion of the nodes and edges in the model, and aiming at each component node [Image Omitted] Based on its original properties [Image Omitted] Deducing functional semantic roles by:

[Image Omitted]

wherein, the [Image Omitted] Representing original properties of a given component [Image Omitted] When the component belongs to the conditional probability distribution of the role c , c represents a predefined component function role; [Image Omitted] Representing the original set of properties of component v , including [Image Omitted] Basic attributes are equal; [Image Omitted] a large language model is represented for semantic reasoning based on the original attributes. Roles with highest confidence [Image Omitted] Storing the confidence value into the node attribute;

normalizing text attribute of component, comprising [Image Omitted] Generating semantically equivalent alias sets for supporting subsequent fuzzy matching, combining structural features of the pages and jump histories, dividing the pages into three types of normal, temporary and cyclic through formulas, [Image Omitted] The logical association and the cross-page function dependence among components which are not directly observed in the inferred exploration process are written into the graph database, and finally the enhanced knowledge graph is formed [Image Omitted] .

The mobile software automation method based on semantic knowledge graph and LLM enhanced reasoning comprises the following steps:

traversing the complete knowledge graph [Image Omitted] If the nodes exceeding the preset value have the same father node, layout structure and role type, only the text and image attributes have difference, and the area supports the scrolling operation, judging the dynamic area [Image Omitted] In the automatic execution process of the execution and monitoring steps, the dynamic instance in the current page is extracted in real time through a system level command [Image Omitted] Instances are matched by structural similarity [Image Omitted] Binding with the corresponding template node T , and calculating the matching score of the instance and the instruction through a formula when the user instruction relates to the dynamic target:

[Image Omitted]

Wherein the method comprises the steps of [Image Omitted] Representing dynamic instances [Image Omitted] Query targets with user instructions [Image Omitted] Matching scores of (2); [Image Omitted] representing text similarity functions for computing instance text [Image Omitted] With query targets [Image Omitted] Is a text overlap degree of (2); [Image Omitted] representing dynamic instances [Image Omitted] Dynamic text properties of (a); [Image Omitted] representing semantic feature similarity functions for computing instance semantic features [Image Omitted] With query targets [Image Omitted] Semantic relatedness of (c); [Image Omitted] representing dynamic instances [Image Omitted] Is a semantic feature set of (1); [Image Omitted] Representing examples [Image Omitted] And (3) giving a descending weight to the position weight in the dynamic area according to the display sequence, and screening the example with the highest score as an interaction target.

The mobile software automation method based on semantic knowledge graph and LLM enhanced reasoning comprises the following steps of:

based on complete knowledge graph [Image Omitted] Searching the interaction target, carrying out multidimensional semantic grounding, and calculating a matching score of the candidate node and the instruction through a formula: [Image Omitted] , wherein, [Image Omitted] Representing candidate nodes retrieved from the knowledge graph; [Image Omitted] Representing a standardized user instruction query target; [Image Omitted] the weight coefficients of text similarity, semantic embedding similarity, role compatibility and context correlation are respectively; selecting the node with the highest score as a grounding target node [Image Omitted] ;

Modeling a task completion process as [Image Omitted] The path searching problem on the node corresponding to the current UI state [Image Omitted] Starting from a grounded target node [Image Omitted] And searching an optimal path formed by legal action edges by adopting a shortest path searching algorithm as an end point, and converting the path into an instruction operation sequence comprising clicking, inputting and sliding operations.

The mobile software automation method based on semantic knowledge graph and LLM enhanced reasoning comprises the following steps:

converting instruction operation sequences into system-level input events, extracting a control tree of a current page through a system-level command after each operation is completed in the execution process, and calculating page signatures [Image Omitted] Signature with expected page [Image Omitted] Comparing, if it meets [Image Omitted] The $\text{sim}()$ represents the page signature similarity function defined before, and the θ represents the state matching threshold value, so that the current operation is judged to be successfully executed, the next action is continuously executed, and otherwise, an exception handling mechanism is triggered;

The exception handling mechanism comprises identifying exception type, if the exception type is a temporary popup window, executing closing operation and then continuing the original sequence, if the exception type is jump failure or dynamic instance change, repeatedly executing current action, wherein the retry times do not exceed a preset threshold value, and the retry failure is based on the result of the retry failure [Image Omitted] The path is re-planned. If the exception type is overtime exception, stopping execution;

and when the current page state meets a matching threshold value with the page state corresponding to the target node and accords with the core intention of the user instruction, judging that the task is successfully executed.

As shown in fig. 7, the present invention further provides a mobile software automation device based on semantic knowledge graph and LLM enhanced reasoning, which includes:

The initial knowledge graph construction module extracts interface information when the mobile equipment runs the mobile application APP, and searches and obtains a unique page signature of each access page according to the content and the function of each access page in the interface information so as to construct an initial knowledge graph [Image Omitted] The node set V comprises page nodes and interaction component nodes thereof, and the edge set E comprises structural edges reflecting father-son inclusion relation and action edges for recording operation jumps;

the semantic enhancement and knowledge enrichment module is used for carrying out initial knowledge graph according to a lightweight large language model LLM and a scene prompt template [Image Omitted] Performing deep semantic completion, deducing functional semantic roles of the nodes according to the attributes of the components by using LLM, normalizing the text of the nodes to generate a group of equivalent semantics for supporting fuzzy matching, combining the initial knowledge graph [Image Omitted] Classifying interface types by LLM to obtain the interface types belonging to normal, temporary or cyclic categories, combining interface types, equivalent semantics and functional semantics roles, and utilizing LLM to infer logical association among components and cross-page function dependence which are not directly observed in exploration so as to complement the initial knowledge graph [Image Omitted] Obtaining a semantically enhanced complete knowledge graph [Image Omitted] ;

Dynamic content processing module, in complete knowledge graph [Image Omitted] Introducing a representation mechanism of template instance separation, and identifying a complete knowledge graph [Image Omitted] As a structural template node [Image Omitted] Recording its fixed layout mode, component role and semantic label, and using specific dynamic item as run-time example [Image Omitted] Contains the text and image characteristics which change in real time and is connected with the corresponding template nodes through a dynamic binding mechanism [Image Omitted] Associating;

the instruction analysis and graph constraint reasoning module receives natural language instructions input by a user [Image Omitted] Analyzing, performing multidimensional semantic grounding, and based on a complete knowledge graph [Image Omitted] Calculating

natural language instructions by integrating multiple dimensions such as text similarity, semantic embedding correlation, component role compatibility and interface context [Image Omitted] Matching score with candidate UI elements [Image Omitted] Selecting candidate UI elements with highest scores as target components, performing graph-based action planning, and modeling a task completion process as in [Image Omitted] The path searching problem searches the shortest path formed by legal action edges by taking the current state node as a starting point and the grounding target node as an end point, and generates an instruction operation sequence;

the execution and monitoring module converts the abstract action sequence of the instruction operation sequence into a system-level input event comprising multi-step operation so as to be automatically executed on the mobile equipment, monitors the change of interface information in the execution process, re-acquires the current control tree after each step of operation, calculates page signature, compares the current control tree with the expected state, and automatically triggers an exception handling mechanism by the module if unexpected popup, jump failure or overtime and the like are detected, and adopts closing popup, operation retry or path re-planning to complete the natural language instruction [Image Omitted] .

The invention also provides an electronic device, which comprises the mobile software automation device based on semantic knowledge graph and LLM enhanced reasoning, wherein the electronic device is connected with an information display device, and the information display device is used for displaying the execution result of the natural language instruction by using display parameters and attributes set by a user or through an artificial intelligent model.

The invention also provides a computer readable storage medium, on which a computer program is stored, which when being executed by a processor, realizes the steps of the mobile software automation method based on semantic knowledge graph and LLM enhanced reasoning.

The invention also provides a computer program product, which comprises a computer program, wherein the computer program realizes the steps of the mobile software automation method based on semantic knowledge graph and LLM enhanced reasoning when being executed by a processor.

The advantages of the invention are as follows:

The deep coordination of the semantic capacity of the large language model and the structural constraint of the knowledge graph is realized, the large language model provides strong understanding capacity for the rapid construction and semantic enrichment of the knowledge graph, the knowledge graph provides a real boundary and a navigation structure for the reasoning process of the large language model, and the end-to-end automatic process from the natural language instruction to the system level operation is ensured to have high precision and high reliability.

The method provided by the invention has high interface coverage rate, high instruction resolution accuracy and reliable task completion rate in real Android application, and provides a solution with clear structure, reusability and easy maintenance for mobile software automation.

Drawings

FIG. 1 is a flow chart of an end-to-end execution of the mobile software automation method of the present invention.

FIG. 2 is a data-centric core data processing chain diagram illustrating how raw UI information is gradually transformed into an executable action sequence under the framework of the present invention.

FIG. 3 is a flowchart of an automated UI exploration for initial knowledge graph construction.

FIG. 4 is a diagram of the core interactions between a user, a large language model, and a mobile terminal device.

FIG. 5 illustrates in detail an exception handling logic diagram during execution of an action sequence.

FIG. 6 is a flow chart of the method of the present invention;

FIG. 7 is a block diagram of the apparatus of the present invention;

FIG. 8 is a schematic diagram of a first electronic device according to the present invention;

FIG. 9 is a schematic diagram of an application environment of a first electronic device according to the present invention;

fig. 10 is a schematic structural diagram of a second electronic device according to the present invention.

Reference numerals:

A-a first electronic device;

B-a mobile software automation device based on semantic knowledge graph and LLM enhanced reasoning;

c-data acquisition equipment;

d-an information display device;

1000-a second electronic device;

an I-calculation unit;

II-ROM;

III-RAM;

IV-bus;

V-interface;

VI-an input unit;

VII-an output unit;

VIII-storage medium;

IX-communication unit.

Detailed Description

It is noted that in the present application, relational terms such as first and second, and the like are used solely to distinguish one entity or action from another entity or action without necessarily requiring or implying any actual such relationship or order between such entities or actions. Also, the terms "comprises," "comprising," or any other variation thereof, are intended to cover a non-exclusive inclusion, such that a process, method, article, or apparatus that comprises a list of elements does not include only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus.

Without further limitation, an element defined by the phrase "comprising one does not exclude the presence of other like elements in a process, method, article, or apparatus that comprises an element.

The processor in the invention is a control center of the electronic equipment, and can be one processor or a collective name of a plurality of processing elements. For example, one or more central processing units (central processing unit, CPU), or Application SPECIFIC INTEGRATED Circuits (ASIC), or one or more integrated circuits configured to implement embodiments of the present invention, such as one or more microprocessors (DIGITAL SIGNAL processors, DSPs), or one or more field programmable gate arrays (fieldprogrammable GATE ARRAY, FPGAs).

In the alternative, the processor may perform various functions of the electronic device by executing or executing software programs stored in memory, and invoking data stored in memory.

In a particular implementation, the processor may include one or more CPUs as one embodiment. Each of these processors may be a single-core processor (single-CPU) or a multi-core processor (multi-CPU). A processor herein may refer to one or more devices, circuits, and

/or processing cores for processing data (e.g., computer program instructions). The electronic device may include a server, desktop computer, notebook computer, smart phone, tablet computer, embedded computer, etc., where the embedded computer includes a carrier, robot, etc.

The memory is configured to store a software program for executing the solution of the present invention, and the processor is used to control the execution of the software program, and the specific implementation manner may refer to the above method embodiment, which is not described herein again.

It should be noted that the structure of the electronic device shown in the drawings of the present invention is not limited thereto, and an actual knowledge structure recognition device may include more or less components than those shown in the drawings, or may combine some components, or may arrange different components.

The above embodiments may be implemented in whole or in part by software, hardware (e.g., circuitry), firmware, or any other combination. When implemented in software, the above-described embodiments may be implemented in whole or in part in the form of a computer program product. The computer program product comprises one or more computer instructions or computer programs. When the computer instructions or computer program are loaded or executed on a computer, the processes or functions described in accordance with embodiments of the present invention are produced in whole or in part. The computer may be a general purpose computer, a special purpose computer, a computer network, or other programmable apparatus. The computer instructions may be stored in a computer-readable storage medium or transmitted from one computer-readable storage medium to another computer-readable storage medium, for example, the computer instructions may be transmitted from one website site, computer, server, or data center to another website site, computer, server, or data center by wired (e.g., infrared, wireless, microwave, etc.). The computer readable storage medium may be any available medium that can be accessed by a computer or a data storage device such as a server, data center, etc. that contains one or more sets of available media. The usable medium may be a magnetic medium (e.g., floppy disk, hard disk, magnetic tape), an optical medium (e.g., DVD), or a semiconductor medium. The semiconductor medium may be a solid state disk.

It should also be understood that the term "and/or" is merely an association relationship describing the associated object, and means that three relationships may exist, for example, a and/or B, and may mean that a exists alone, while a and B exist alone, and B exists alone, wherein a and B may be singular or plural. In addition, the character "/" herein generally indicates that the associated object is an "or" relationship, but may also indicate an "and/or" relationship, and may be understood by referring to the context.

In the present invention, "at least one" means one or more, and "a plurality" means two or more. "at least one of" or the like means any combination of these items, including any combination of single item(s) or plural items(s). For example, at least one (a, b, or c) of a, b, c, a-b, a-c, b-c, or a-b-c may be represented, wherein a, b, c may be single or plural.

It should also be understood that, in various embodiments of the present invention, the sequence numbers of the foregoing processes do not mean the order of execution, and the order of execution of the processes should be determined by the functions and internal logic thereof, and should not constitute any limitation on the implementation process of the embodiments of the present invention.

In the several embodiments provided by the present invention, it should be understood that the disclosed apparatus, devices, and methods may be implemented in other ways. For example, the apparatus embodiments described above are merely illustrative, e.g., the division of the units is merely a logical function division, and there may be additional divisions when actually implemented, e.g., multiple units or components may be combined or integrated into another device, or some features may be omitted or not performed. Alternatively, the coupling or direct coupling or communication connection shown or discussed with each other may be an indirect coupling or communication connection via some interfaces, devices or units, which may be in electrical, mechanical or other form.

The units described as separate units may or may not be physically separate, and units shown as units may or may not be physical units, may be located in one place, or may be distributed on a plurality of network units. Some or all of the units may be selected according to actual needs to achieve the purpose of the solution of this embodiment.

In addition, each functional unit in the embodiments of the present invention may be integrated in one processing unit, or each unit may exist alone physically, or two or more units may be integrated in one unit.

The functions, if implemented in the form of software functional units and sold or used as a stand-alone product, may be stored in a computer-readable storage medium. Based on this understanding, the technical solution of the present invention may be embodied essentially or in a part contributing to the prior art or in a part of the technical solution, in the form of a software product stored in a storage medium, comprising several instructions for causing a computer device (which may be a personal computer, a server, a network device, etc.) to perform all or part of the steps of the method according to the embodiments of the present invention. The storage medium includes a

U disk, a removable hard disk, a read-only memory (ROM), a random access memory (random access memory, RAM), a magnetic disk, an optical disk, or other various media capable of storing program codes.

The invention provides a mobile software autonomous automation method based on structured semantic knowledge graph and LLM enhanced reasoning, which is characterized in that a collaborative framework of automatic UI exploration and structured modeling, semantic knowledge graph enhancement, constraint reasoning and planning-robust execution monitoring is adopted, so that the technical pain points of poor interface variability adaptability, inaccurate analysis of natural language instructions, difficult dynamic content processing, difficult multiplexing of automated assets and the like in the existing mobile automation technology are solved, and the mobile terminal autonomous automation with high precision, high robustness and maintainability is realized.

The automatic control method comprises a plurality of modules, wherein each module performs logic collaborative work of closed loop execution according to data acquisition, knowledge construction, semantic enhancement, constraint reasoning, instruction analysis.

Capturing UI interface information (xml file) of an application through ADB (Android Debug Bridge) technology, inputting interface information of a mobile application APP in real time when running into an automatic UI exploration and initial knowledge graph construction module, and executing a series of actions such as input clicking and the like through ADB to realize system-level interaction instruction driven automatic exploration. The exploration process adopts a mixed traversal strategy and combines breadth-first and depth-first traversal to cover the navigation skeleton and deep content of the application. And extracting a control tree of each accessed page, calculating a unique page signature based on the layout structure, the text content and the component type, and realizing the duplicate removal identification of the interface state and the filtration of the temporary page. Constructing an initial UI knowledge graph based on the exploration result [Image Omitted] The node set V comprises page nodes and interactable component nodes, and the edge set E comprises structural edges reflecting parent-child inclusion relations and action edges for recording operation jumps. Specifically, the layout of each page with specific functions is abstracted into a node, for example, an icon of an app exists in the page, and the layout can be abstracted into a corresponding page node.

The semantic enhancement and knowledge enrichment module receives an initial knowledge graph [Image Omitted] And introducing a lightweight large language model and a scenerization prompt template, and performing deep semantic completion on the map. The module uses LLM to infer its functional semantic role from the text, type, and location of the component, e.g., identify it as a "search box", "submit button", or "return icon". At the same time, it normalizes the UI text and generates a set of semantically equivalent aliases to support subsequent fuzzy matching. The module also combines the page structure and the jump history thereof, and classifies the whole interactive interface by LLM, and judges that the interactive interface belongs to the category of normal, temporary or circulation. In addition, it can infer inter-component logical associations and cross-page functional dependencies that are not directly observed in the exploration, thus complementing knowledge relationships. Finally, the module outputs a complete knowledge graph subjected to comprehensive semantic enhancement [Image Omitted] .

The dynamic content template-instance abstraction module is dedicated to handling dynamically generated list, card, etc. content in an application. The module introduces a template-instance separated representation mechanism into the knowledge graph, and the graph only stores structural template nodes of the dynamic region [Image Omitted] The fixed layout patterns, component roles and semantic tags are recorded. Specific dynamic items are then used as runtime instances [Image Omitted] Features such as text, images and the like which change in real time are contained and are associated with the corresponding template nodes T through a dynamic binding mechanism, namely, in a dynamic area, an instance (dynamic item) is changeable, can be rebindd in each update, and can be deleted without existence. The design effectively ensures the compactness and stability of the knowledge graph and simultaneously ensures the accurate positioning and interactive operation of dynamic content.

The dynamic area refers to that text content (dynamic item) in the area can be changed, but functions in the area are unchanged, such as a contact list of social software APP, and each click-in is that a message page can send a message, namely the same function but different specific contents. There is also entertainment news software APP, the specific content of each news is different, but the presentation functions are the same. The area it occupies in the page is unchanged, so this area is a dynamic area.

The instruction analysis and graph constraint reasoning module receives natural language instructions input by a user [Image Omitted] For example, the user speaks "click next song", and the text sequence of this sentence is a natural language instruction. The analysis process mainly comprises two steps of firstly carrying out multi-dimensional semantic grounding and based on semantic knowledge graph [Image Omitted] The system integrates multiple dimensions such as text similarity, semantic embedding correlation, component role compatibility, interface context and the like, calculates instructions and candidate UI elements (complete knowledge graph [Image Omitted] Middle node), the calculation formula is: [Image Omitted] , wherein, [Image Omitted] Representing the multi-dimensional matching total score of the candidate node c and the query target q, wherein c represents the candidate node retrieved from the knowledge graph, and q represents the normalized user instruction query target; [Image Omitted] the weight coefficients of text similarity, semantic embedding similarity, role compatibility and context correlation are respectively; [Image Omitted] representing a text and alias similarity function, and calculating the

overlapping degree of the text/alias of the candidate node c and q ; [Image Omitted] representing a semantic embedded cosine similarity function, and calculating semantic vector correlation of candidate nodes c and q ; [Image Omitted] Representing a role compatibility function, and evaluating the intention matching degree of the function roles and q of the candidate node c ; [Image Omitted] And (5) representing a context correlation function and measuring the degree of correlation between the page where the candidate node c is positioned and the current navigation context. The target assembly or template pointed by the instruction is precisely locked, then the drawing base action planning is carried out, and the task completion process is modeled as in [Image Omitted] The path searching problem searches the shortest path formed by legal action edges by taking the current state node as a starting point and the grounding target node as an end point, and generates a series of abstract operation sequences such as clicking, inputting, sliding and the like. For example, a social APP exists in the current interface, the current state node is the current page node, the node is connected with the social APP component node through a relation edge (including), the user instruction requests to enter a chat interface with a small opening, and the small opening in the contact is the final grounding target node.

The robust execution and status monitoring module is responsible for converting the planned abstract action sequence into specific system-level input events, such as simulated touch, text injection, etc., and executing on the mobile device through the ADB or barrier-free service interface. In the execution process, the module continuously monitors the interface state change, and re-acquires the current control tree after each operation, calculates the page signature and compares the current control tree with the expected state. If unexpected popup windows, jump failures or overtime conditions are detected, the module automatically triggers an exception handling mechanism, such as closing the popup windows, operating retries or path re-planning, thereby ensuring reliable completion of tasks in a dynamic environment.

In addition, the system introduces a knowledge graph increment maintenance module as an optional component, continuously feeds back newly discovered interface states, components and interaction relations in an automatic execution process and updates the newly discovered interface states, components and interaction relations to the semantic knowledge graph [Image Omitted] The online learning and incremental expansion of the atlas are realized, so that the system has the capability of continuous evolution.

To ensure the initial knowledge graph [Image Omitted] Pruning and optimizing UI information (information in an xml file) after UI exploration are needed, and specific measures comprise integrating same-level fragmentation information through text merging, transmitting a dynamic state through an attribute inheritance mechanism, and clearing invalid nodes with empty text, messy codes and abnormal coordinates, so that the information density and quality of a map are improved.

Firstly, executing invalid node clearing operation, traversing a component node set generated by UI exploration, checking text attribute context attribute and bounds attribute of nodes one by one, and if the text attribute and the context attribute are simultaneously blank character strings or respectively contain unresolved messy code characters, or bounds attribute (coordinate range) exceeds the resolution range of the screen of the current equipment, is of a negative number or non-numerical type or is full-screen, marking the node as an invalid node, and then removing all the component nodes marked as invalid from the node set of the initial knowledge graph. After the invalid node is cleared, the same-level fragmentation information is integrated through text merging, a UI control tree is traversed, text leaf nodes under the same father node are identified), the logic sequence of text splicing is determined according to the arrangement sequence of the text nodes in the control tree or the horizontal/vertical layout sequence of coordinates of the text nodes, text attributes and context attributes of the fragmented text nodes are spliced into complete text strings according to the sequence, new text component nodes are created to replace the original fragmented nodes, meanwhile, the hierarchical association of the text component nodes with the father node is maintained, and the original fragmented nodes are removed. And if the dynamic attribute of the child node is clickable, enabled, inheriting the default dynamic state from the same-level and same-type component nodes if the dynamic attribute of the child node is unknown or not clear, and if the father node has an associated dynamic state rule, transmitting the rule to the child node to complement the dynamic state attribute of the child node.

The key technical point of the invention is that the deep coordination of the semantic capacity of the large language model and the structural constraint of the knowledge spectrum is realized, the large language model provides strong understanding capacity for the rapid construction and semantic enrichment of the knowledge spectrum, the knowledge spectrum provides a real boundary and a navigation structure for the reasoning process of the large language model, and the end-to-end automatic process from the natural language instruction to the system level operation is ensured to have high precision and high reliability.

In order to make the above features and effects of the present invention more clearly understood, the following specific examples are given with reference to the accompanying drawings. The present specification discloses one or more embodiments that incorporate the features of the invention. The disclosed embodiments are merely illustrative. The scope of the invention is not limited to the disclosed embodiments, but is defined by the appended claims.

In summary, the invention provides a mobile software automation method based on structured semantic knowledge graph and LLM enhanced reasoning, which converts mobile terminal interface information with dynamic variability and lack of unified characterization into a structured knowledge graph through automatic UI exploration and semantic modeling, so that a large language model can accurately grasp

semantic roles, page navigation relations and interaction logics of UI components, and simultaneously improves the integrity and reliability of knowledge representation through optimization means such as page deduplication, semantic enrichment, relation complementation and the like. The specific construction and process flow of the automated method will be described in detail below in conjunction with fig. 1-5.

FIG. 1 shows a semantic UI knowledge graph based on the present invention [Image Omitted] The whole implementation flow chart of the mobile software autonomous automation method for enhancing reasoning clearly presents a whole flow closed loop from interface information acquisition, knowledge modeling and semantic enhancement to instruction execution, and intuitively displays the cooperative logic and data flow relation of six functional modules. FIG. 1 clearly presents the overall process closed loop from initial knowledge graph construction, semantic completion and knowledge enrichment, dynamic content-instance abstraction, user instruction resolution to final instruction execution. In the figure, the data flow and the cooperative logic between six functional modules are definitely identified through arrows and module boxes, and the data flow and the cooperative logic comprise an automatic UI exploration and initial knowledge graph construction module, a semantic enhancement and knowledge enrichment module, a dynamic content template-instance abstraction module, an instruction analysis and constraint reasoning module and a robust execution and state monitoring module. The figure intuitively embodies the integrated technical route of data acquisition-knowledge construction-semantic enhancement-constraint reasoning-instruction analysis-execution monitoring.

Fig. 3 illustrates a complete exploration link from exploration initialization, page dequeuing, control tree extraction, page signature calculation, deduplication judgment, operable action enumeration and execution, to temporary page filtering, skip relation recording, new page enqueueing, and intuitively presents the construction process of an initial knowledge graph under a hybrid traversal strategy.

As shown in fig. 3, the interface information of the mobile application in running first enters an automatic UI exploration and initial knowledge graph construction module, and the core objective is to convert the interface information without unified structure into normalized initial knowledge graph [Image Omitted] Wherein the node sets [Image Omitted] Edge set [Image Omitted] . When the program runs, firstly, the exploration initialization is carried out, and the application initial starting page is started [Image Omitted] Inclusion exploration queue [Image Omitted] I.e. [Image Omitted] Initializing a set of accessed page signatures [Image Omitted] Explicit action enumeration types include click, text entry, slide, system return four basic interactions. The exploration process adopts a mixed traversal strategy and combines breadth-first and depth-first traversal, firstly, the navigation skeleton of the overlay application is traversed through breadth-first traversal, and then the deep component structure in the page is mined through depth-first traversal.

For each page in the queue [Image Omitted] The control tree is extracted by a system level command [Image Omitted] A unique page signature is calculated based on the layout structure, text content and component type, which is defined as:

[Image Omitted]

wherein, the [Image Omitted] Representing pages [Image Omitted] The unique structure signature of the (a) is used for page duplication removal and identity recognition; [Image Omitted] representing pages [Image Omitted] The text token set of (2) is generated by normalizing visible component text; [Image Omitted] representing pages [Image Omitted] By mapping control trees to layout hash values of (a) [Image Omitted] Performing hash calculation to obtain the data for capturing the parent-child hierarchical relationship and ordering characteristics of the components; [Image Omitted] representing pages [Image Omitted] Is extracted from the component attributes and is used for representing the component function category determined after LLM judgment.

Calculating the similarity of the page signatures through a weighted combination formula, and judging whether the page has been accessed or not:

[Image Omitted]

wherein, the [Image Omitted] Representing pages [Image Omitted] Is signed by (a) [Image Omitted] And page [Image Omitted] Is signed by (a) [Image Omitted] The similarity between the two is in the value range of 0, 1; [Image Omitted] the weight coefficients of the text similarity, the layout consistency and the role similarity respectively satisfy the following conditions [Image Omitted] ; [Image Omitted] Representing a Jacquard similarity function for calculating the degree of overlap of the two sets; [Image Omitted] Respectively are pages [Image Omitted] A text token set of (a); [Image Omitted] Respectively are pages [Image Omitted] $1[-J]$ is an indication function, returns 1 when the condition in brackets is satisfied, otherwise returns 0; [Image Omitted] Respectively are pages [Image Omitted] Is a component role set of (1). If the similarity reaches a preset threshold, determining that the page is repeated, skipping the subsequent processing, otherwise, adding the page signature into the accessed set [Image Omitted] .

Enumerating an operational action set for a current page [Image Omitted] Executing actions one by one [Image Omitted] Observing the new page generated [Image Omitted] Face, identify the temporary pages such as popup window, permission prompt, etc. through heuristic rules,

if [Image Omitted] Fewer number of operable components, cancellation, permission of feature text, etc., and return to the original page after performing the return action [Image Omitted] Judging that the page is a temporary page, not recording the jump relation, otherwise recording the jump relation [Image Omitted]) And will [Image Omitted] Enqueue [Image Omitted] Executing a system return action back to the page [Image Omitted] 。

Constructing an initial knowledge graph based on the exploration result [Image Omitted] For all the nodes corresponding to the unique pages, the page signature is taken as the unique identification, [Image Omitted] For visible and interactable component nodes in the control tree, each component node satisfies:

[Image Omitted]

Wherein the method comprises the steps of [Image Omitted] The text is displayed for the component and, [Image Omitted] Is a unique identification for the component and, [Image Omitted] For a range of coordinates of the component, [Image Omitted] The interaction attributes of the components are respectively, [Image Omitted] Is a component type attribute.

In order to improve the quality of the initial graph, pruning and optimization are needed, namely invalid nodes with empty texts, messy codes and abnormal coordinates are removed, the same-level fragmentation information is integrated through text merging, adjacent text components under the same father node are integrated according to page display sequence, dynamic states are transferred through an attribute inheritance mechanism, the integrity of component attributes is ensured, and finally, the initial knowledge graph with clear structure and high information density is formed [Image Omitted] 。

Initial knowledge graph [Image Omitted] After the construction is completed, the semantic enhancement and knowledge enrichment module is input, and the core goal of the module is to complement the semantic attribute and the logic relation of the atlas by means of the semantic understanding capability of the large language model to generate an enhanced knowledge atlas [Image Omitted] 。

And introducing a lightweight large language model and a scene prompt template, and carrying out semantic completion on nodes and edges in the initial map. For each component node [Image Omitted] Based on its original properties [Image Omitted] Deducing functional semantic roles through formulas:

[Image Omitted]

wherein, the [Image Omitted] Representing original properties of a given component [Image Omitted] When the component belongs to the conditional probability distribution of the role c , c represents a predefined component function role; [Image Omitted] Representing the original set of properties of component v , including [Image Omitted] Basic attributes are equal; [Image Omitted] a large language model is represented for semantic reasoning based on the original attributes. Roles with highest confidence [Image Omitted] The confidence value is stored in the node attribute.

Normalizing text attribute of component, comprising [Image Omitted] Canonical _label and semantically equivalent alias sets are generated for supporting subsequent fuzzy matching. Combining the structural characteristics of the page and the jump history, dividing the page into three types of normal, temporary and cyclic through a formula, [Image Omitted] Logical association and cross-page function dependence among components which are not directly observed in the inferred exploration process are also written into a graph database, and finally an enhanced knowledge graph which is rich in semantics and can be efficiently queried according to roles, texts and relationship types is formed [Image Omitted] 。

Fig. 2 specifically depicts a complete data conversion process of sequentially performing initial knowledge graph construction, knowledge completion and semantic summarization driven by a large language model, knowledge graph prefix tree construction, graph constraint decoding process, and analysis of a user natural language instruction from the original mobile terminal page input, and finally generating a structured action instruction sequence. As shown in fig. 2, the dynamic content template-instance abstract module specifically processes the contents such as a list and a card generated dynamically in the application, and the core objective is to realize accurate interaction on the dynamic content while ensuring compactness and stability of the knowledge graph through a template-instance separation mechanism.

Traversing [Image Omitted] If the plurality of components have the same father node, layout structure and role type, only the attributes such as text, image and the like have differences, and the area supports the scrolling operation, the dynamic area is determined. At this time, a template node T is created for each dynamic region, and the template node is stored [Image Omitted] And deleting repeated example nodes in the dynamic region, and only reserving template nodes to control the scale of the map. In the automatic execution stage, dynamic

examples in the current page are extracted in real time through system level commands [Image Omitted] Instances are matched by structural similarity [Image Omitted] Binding with the corresponding template node T. When a user instruction relates to a dynamic target, calculating a matching score of an instance and the instruction through a formula:

[Image Omitted]

Wherein the method comprises the steps of [Image Omitted] Representing dynamic instances [Image Omitted] Query targets with user instructions [Image Omitted] The higher the score, the higher the matching degree; [Image Omitted] representing text similarity functions for computing instance text [Image Omitted] With query targets [Image Omitted] Is a text overlap degree of (2); [Image Omitted] representing dynamic instances [Image Omitted] Dynamic text properties of (a); [Image Omitted] representing semantic feature similarity functions for computing instance semantic features [Image Omitted] With query targets [Image Omitted] Semantic relatedness of (c); [Image Omitted] representing dynamic instances [Image Omitted] Is a semantic feature set of (1); [Image Omitted] Representing examples [Image Omitted] And (3) giving a descending weight to the position weight in the dynamic area according to the display sequence, and screening the example with the highest score as an interaction target.

The core objective of the instruction analysis and graph constraint reasoning module is to convert a natural language instruction I input by a user into an executable abstract action sequence, the module preprocesses the natural language instruction I, and extracts core intention and key parameters through word segmentation and semantic normalization operation to convert the spoken language expression into standardized query.

Based on enhanced knowledge graph [Image Omitted] Searching candidate nodes which are matched with the instruction intention, carrying out multidimensional semantic grounding, and calculating a matching score of the candidate nodes and the instruction through a formula: [Image Omitted] , wherein, [Image Omitted] Representing candidate nodes retrieved from the knowledge graph; [Image Omitted] Representing a standardized user instruction query target; [Image Omitted] the weight coefficients of text similarity, semantic embedding similarity, role compatibility and context correlation are respectively. Selecting the node with the highest score as a grounding target node [Image Omitted] .

Modeling a task completion process as [Image Omitted] The path searching problem on the node corresponding to the current UI state [Image Omitted] Starting from a grounded target node [Image Omitted] And searching an optimal path formed by legal action edges by adopting a shortest path searching algorithm for an end point, converting the path into an abstract action sequence comprising operations such as clicking, inputting, sliding and the like, and determining a target component, an interaction type and parameters of each operation.

FIG. 4 illustrates a user inputting a natural language instruction, the instruction is parsed and planned via an integrated graph constraint inference capability large language model to generate an executable sequence of operations, which ultimately drives an automated execution of a complete interactive link on a mobile device.

Fig. 5 illustrates in detail the closed loop links from the acquisition of the action sequence, the execution of the current action, the monitoring of the interface state, the status determination, the anomaly identification and the classification processing, the limitation of the retry times and the circulation relationship of each link are clarified, and the robustness of the execution process is ensured. As shown in FIG. 5, the instruction execution and status monitoring module is responsible for converting the abstract action sequence into system-level operation and executing, and simultaneously monitoring the interface status in real time, handling abnormal situations and ensuring the reliable completion of tasks. Firstly, converting an abstract action sequence into a specific system-level input event, extracting a control tree of a current page through a system-level command after each operation is completed in the execution process, and calculating a page signature [Image Omitted] Signature with expected page [Image Omitted] Comparing, if it meets [Image Omitted] And judging that the current operation is successfully executed, continuing to execute the next action, and otherwise, triggering an exception handling mechanism. When the abnormal state is detected, firstly, the abnormal type is identified, if the abnormal type is the temporary popup window, the closing operation is executed, and then the original sequence is continued. If the jump fails or the dynamic instance changes, the current action is repeatedly executed, the retry times do not exceed the preset threshold, and the retry failure is based on [Image Omitted] The path is re-planned. If the time-out is abnormal, stopping execution and feeding back the abnormal reason. And when the current page state meets a matching threshold value with the page state corresponding to the target node and accords with the core intention of the user instruction, judging that the task is successfully executed. Otherwise, judging failure, and outputting failure reasons and manual intervention suggestions.

The knowledge graph increment maintenance module is used as an optional component, the core aim is to realize online learning and increment expansion of the graph, so that the system can adapt to iterative updating of an application UI, the interface state is continuously monitored in the automatic execution process, and if the condition that the interface state is not incorporated is detected [Image Omitted] Automatically triggering incremental exploration flow, namely extracting a new page repetition control tree, calculating page signatures, de-

duplication judging, executing a semantic enhancement flow on a new component, and supplementing new nodes and new relations to the new component [Image Omitted] 。

The foregoing description of the exemplary embodiments of the invention is not intended to limit the invention to the particular embodiments disclosed, but on the contrary, the intention is to cover all modifications, equivalents, and alternatives falling within the spirit and scope of the invention.

The following is a system example corresponding to the above method example, and this embodiment mode may be implemented in cooperation with the above embodiment mode. The related technical details mentioned in the above embodiments are still valid in this embodiment, and in order to reduce repetition, they are not repeated here. Accordingly, the related technical details mentioned in the present embodiment can also be applied to the above-described embodiments.

The invention also provides a mobile software automation device based on semantic knowledge graph and LLM enhanced reasoning, which comprises:

The initial knowledge graph construction module extracts interface information when the mobile equipment runs the mobile application APP, and searches and obtains a unique page signature of each access page according to the content and the function of each access page in the interface information so as to construct an initial knowledge graph [Image Omitted] The node set V comprises page nodes and interaction component nodes thereof, and the edge set E comprises structural edges reflecting father-son inclusion relation and action edges for recording operation jumps;

the semantic enhancement and knowledge enrichment module is used for carrying out initial knowledge graph according to a lightweight large language model LLM and a scene prompt template [Image Omitted] Performing deep semantic completion, deducing functional semantic roles of the nodes according to the attributes of the components by using LLM, normalizing the text of the nodes to generate a group of equivalent semantics for supporting fuzzy matching, combining the initial knowledge graph [Image Omitted] Classifying interface types by LLM to obtain the interface types belonging to normal, temporary or cyclic categories, combining interface types, equivalent semantics and functional semantics roles, and utilizing LLM to infer logical association among components and cross-page function dependence which are not directly observed in exploration so as to complement the initial knowledge graph [Image Omitted] Obtaining a semantically enhanced complete knowledge graph [Image Omitted] ;

Dynamic content processing module, in complete knowledge graph [Image Omitted] Introducing a representation mechanism of template instance separation, and identifying a complete knowledge graph [Image Omitted] As a structural template node [Image Omitted] Recording its fixed layout mode, component role and semantic label, and using specific dynamic item as run-time example [Image Omitted] Contains the text and image characteristics which change in real time and is connected with the corresponding template nodes through a dynamic binding mechanism [Image Omitted] Associating;

the instruction analysis and graph constraint reasoning module receives natural language instructions input by a user [Image Omitted] Analyzing, performing multidimensional semantic grounding, and based on a complete knowledge graph [Image Omitted] Calculating natural language instructions by integrating multiple dimensions such as text similarity, semantic embedding correlation, component role compatibility and interface context [Image Omitted] Matching score with candidate UI elements [Image Omitted] Selecting candidate UI elements with highest scores as target components, performing graph-based action planning, and modeling a task completion process as in [Image Omitted] The path searching problem searches the shortest path formed by legal action edges by taking the current state node as a starting point and the grounding target node as an end point, and generates an instruction operation sequence;

the execution and monitoring module converts the abstract action sequence of the instruction operation sequence into a system-level input event comprising multi-step operation so as to be automatically executed on the mobile equipment, monitors the change of interface information in the execution process, re-acquires the current control tree after each step of operation, calculates page signature, compares the current control tree with the expected state, and automatically triggers an exception handling mechanism by the module if unexpected popup, jump failure or overtime and the like are detected, and adopts closing popup, operation retry or path re-planning to complete the natural language instruction [Image Omitted] 。

As shown in fig. 8, the present invention further provides a first electronic device a, which includes the mobile software automation device based on semantic knowledge graph and LLM enhanced reasoning.

As shown in fig. 9, the first electronic device a may also be connected to a data acquisition device C and an information display device D through a wired or wireless information transmission scheme, where the data acquisition device C is configured to acquire a natural language instruction, for example, "change next song", "send XX information to contact person's mins", and the information display device D is configured to display an execution result of the natural language instruction obtained by executing the present invention.

The information display device D may sort and process the data output by the first electronic device a based on the information display mechanism, so as to improve the readability of the data output by the first electronic device a. The information display mechanism may be preset manually, for example, the data output by the first electronic device a may be displayed visually, and the display parameter may be, for example, a display data range according to a display parameter and/or an attribute set by a user, and the display attribute may be, for example, a display font, a color, whether to scroll and play, etc. The appointed key information is presented to the user, the user can know the information more timely without visiting a secondary page or scrolling the page, and the operation of the user is saved. Or the information display mechanism can be an artificial intelligent AI display model, which can learn the key attention information of the user according to the previous use habits of the user, such as watching time length, clicking times, editing times and the like, so that the automatic user presents rich and necessary key information.

The invention also provides a computer program product, which comprises a computer program, the computer program can be stored on a readable storage medium, and when the computer program is executed by a processor, the computer can execute the semantic knowledge graph and LLM enhanced reasoning method provided by the methods.

The invention also provides a storage medium VIII for storing a computer program for executing the semantic knowledge graph and LLM enhanced reasoning method. It should be understood that the storage medium in embodiments of the present invention may be volatile memory or nonvolatile memory, or may include both volatile and nonvolatile memory. The nonvolatile memory may be a read-only memory (ROM), a Programmable ROM (PROM), an erasable programmable ROM (erasable PROM), an electrically erasable programmable EPROM (EEPROM), or a flash memory. The volatile memory may be random access memory (random access memory, RAM) which acts as external cache memory. By way of example, and not limitation, many forms of random access memory (random access memory, RAM) are available, such as static random access memory (STATIC RAM, SRAM), dynamic Random Access Memory (DRAM), synchronous Dynamic Random Access Memory (SDRAM), double data rate synchronous dynamic random access memory (double DATA RATE SDRAM, DDR SDRAM), enhanced synchronous dynamic random access memory (ENHANCED SDRAM, ESDRAM), synchronous link dynamic random access memory (SYNCHLINK DRAM, SLDRAM), and direct memory bus random access memory (direct rambus RAM, DR RAM).

Fig. 10 shows a schematic block diagram of a second electronic device 1000 that may be used to implement an embodiment of the invention. Second electronic device 1000 the electronic device is intended to represent various forms of digital computers, such as laptops, desktops, personal digital assistants, servers, blade servers, mainframes, and other appropriate computers. The second electronic device 1000 may also represent various forms of mobile devices, such as personal digital processing, cellular telephones, smartphones, wearable devices, and other similar computing devices. The components shown herein, their connections and relationships, and their functions, are meant to be exemplary only, and are not meant to limit implementations of the inventions described and/or claimed herein. The second electronic device 1000 may be the same as or different from the first electronic device a.

The second electronic device 1000 comprises a computing unit I that can perform various suitable actions and processes according to a computer program stored in a read only memory II (ROM) or a computer program loaded from a storage medium VIII into a Random Access Memory (RAM) III. In RAM III, various programs and data required for the operation of the device 1000 may also be stored. The computing units I, ROM II and RAM III are connected to each other via a bus IV. An input/output (I/O) interface V is also connected to the bus IV.

The components in the second electronic device 1000 are connected to an I/O interface V, comprising an input unit VI, e.g. a keyboard, a mouse, etc., an output unit VII, e.g. various types of displays, loudspeakers, etc., a storage medium VIII, e.g. a magnetic disk, an optical disk, etc., and a communication unit IX, e.g. a network card, a modem, a wireless communication transceiver, etc. The communication unit IX allows the second electronic device 1000 to exchange information/data with other devices via a computer network, such as the internet, and/or various telecommunication networks.

The computing unit I may be a variety of general and/or special purpose processing components having processing and computing capabilities. Some examples of computing unit I include, but are not limited to, a Central Processing Unit (CPU), a Graphics Processing Unit (GPU), various dedicated Artificial Intelligence (AI) computing chips, various computing units running machine learning model algorithms, a Digital Signal Processor (DSP), and any suitable processor, controller, microcontroller, etc. The computing unit I performs the respective methods and processes described above, for example the method steps S1-S5. For example, in some embodiments, the method may be implemented as a computer software program tangibly embodied on a machine-readable medium, such as storage medium VIII. In some

embodiments, part or all of the computer program may be loaded and/or installed onto the device 1000 via the ROM II and/or the communication unit IX. When the computer program is loaded into RAM III and executed by the computing unit I, one or more steps of the method described above may be performed. Alternatively, in other embodiments, the computing unit I may be configured to perform the method by any other suitable means (e.g. by means of firmware).

Although embodiments of the present invention have been disclosed above, it is not limited to the details and embodiments shown and described, it is well suited to various fields of use for which the invention would be readily apparent to those skilled in the art, and accordingly, the invention is not limited to the specific details and illustrations shown and described herein, without departing from the general concepts defined in the claims and their equivalents.

Document Claims

Language: (Chinese)

1.一种基于语义知识图谱与LLM增强推理的移动软件自动化方法，其特征在于，包括：

初始知识图谱构建步骤，提取移动设备运行移动应用APP时的界面信息，根据界面信息 中每个访问的页面的内容和功能，探索得到每个访问页面的唯一页面签名，以构建初始知 识图谱 [Image Omitted]，其中节点集V包含页面节点与其交互组件节点，边集E包含反映 父子关系的结构边与记录操作跳转的动作边；

语义增强与知识富集步骤，根据轻量级大语言模型LLM及场景化提示模板，对初始知识 图谱 [Image Omitted] 进行深度语义补全，利用LLM，根据组件的属性推断节点的功能语义角色，并对 节点的文本进行归一化处理，生成一组等效语义，以支撑的模糊匹配；结合初始知识图谱 [Image Omitted] 的结构，由LLM对界面类型进行分类，得到其属于正常、临时或循环类别；结合界面 类型、等效语义、功能语义角色，利用LLM推断在探索中未直接观察到的组件间逻辑关联及 跨页面功能依赖，从而补全该初始知识图谱 [Image Omitted] 的知识关系，得到语义增强的完整知识 图谱 [Image Omitted]；

动态内容处理步骤，在完整知识图谱 [Image Omitted] 中引入模板实例分离的表示机制，识别 完整知识图谱 [Image Omitted] 的动态区域，作为结构模板节点 [Image Omitted]，记录其固定的布局模式、组件 角色和语义标签；具体的动态项作为运行时实例 [Image Omitted]，包含实时变化的文本、图像特征，并通 过动态绑定机制与相应模板节点 [Image Omitted] 关联；

指令解析与图约束推理步骤，接收用户输入的自然语言指令 [Image Omitted]，并进行解析，进行多维 度语义接地，基于完整知识图谱 [Image Omitted]，综合文本相似度、语义嵌入相关性、组件角色兼 容性及界面上下文等多个维度，计算自然语言指令 [Image Omitted] 与候选UI元素的匹配得分 [Image Omitted]，选择得分最高的 候选UI元素作为目标组件；进行图基动作规划，将任务完成过程建模为在 [Image Omitted] 上的路 径搜索问题，以当前状态节点为起点、接地目标节点为终点，搜索由合法动作边构成的最短 路径，并生成指令操作序列；

执行与监控步骤，将该指令操作序列的抽象动作序列转换为包括多步操作的系统级输 入事件，以在该移动设备上自动执行，执行过程中，监 控界面信息变化，在每一步操作后重 新获取当前控件树、计算页面签名，并与预期状态进行比对；若检测到非预期弹窗、跳转失 败或超时等情况，模块自动触发异常处理机制，采取关闭弹窗、操作重试或路径重新规划， 以完成该自然语言指令 [Image Omitted]。

2.如权利要求1所述的基于语义知识图谱与LLM增强推理的移动软件自动化方法，其特征在于，执行与监控步骤包括：

知识图谱增量维护步骤，在该自动执行过程中将新发现的界面状态、组件及交互关系 持续反馈并更新至完整知识图谱 [Image Omitted]，实现 图谱的在线学习与增量扩展，使系统具备 持续进化的能力。

3.如权利要求1所述的基于语义知识图谱与LLM增强推理的移动软件自动化方法，其特 征在于，初始知识图谱构建步骤包括：探索界面信息中 每个访问的页面的内容和功能后，为 确保初始知识图谱 [Image Omitted] 的准确性与有效性，对界面信息进行剪枝，包括通过文本合并整 合同 层级信息、通过属性继承机制传递动态状态，并清除空文本、乱码及坐标异常的无效节 点。

4.如权利要求1所述的基于语义知识图谱与LLM增强推理的移动软件自动化方法，其特征在于，初始知识图谱构建步骤包括：

初始知识图谱 [Image Omitted] 中节点集 [Image Omitted]，边集 [Image Omitted]，[Image Omitted] 为结构边，[Image Omitted] 是动作 边，APP程序运行时，将应用初始启动页面 [Image Omitted] 纳入探索队列 [Image Omitted]，即 [Image Omitted]，初始化已访问页面签名集 合 [Image Omitted]，动作边类型包括点击、文本输入、滑动、 返回操作；

对队列 [Image Omitted] 中的每个页面 [Image Omitted]，执行如下操作：通过系统级命令提取控件树 [Image Omitted]，基于布局 结构、文 本内容与组件类型计算该唯一页面签名：

[Image Omitted]

其中，[Image Omitted] 表示页面 [Image Omitted] 的唯一结构签名，用于页面去重与身份识别；[Image Omitted] 表示页面 [Image Omitted] 的文本token集合，由可见组件文本经规范化处理后生成；[Image Omitted] 表示页面 [Image Omitted] 的布局哈希值，通过对控件树 [Image Omitted] 进行哈希计算得到，用于捕捉组件父子层级关系与排序特征；[Image Omitted] 表 示页面 [Image Omitted] 的组件角色集合，从组件 属性中提取并经LLM判断后确定的用于表征组件功能类 别；

通过加权组合公式计算页面签名相似度，判断页面是否已访问：

[Image Omitted]

其中，[Image Omitted] 表示页面 [Image Omitted] 的签名 [Image Omitted] 与页面 [Image Omitted] 的签名 [Image Omitted] 之间的相似度，取值范围为 [0,1]；[Image Omitted] 分别为文本相似度、布局一致性、角色相似度的权重系数，满足 [Image Omitted]；[Image Omitted] 表示杰卡德相似度函数，用于计算两个集合的重叠程度；[Image Omitted] 分别为页面 [Image Omitted] 的文本token集合；[Image Omitted] 分别为页面 [Image Omitted] 的布局哈希值；1[.]为指示函数，当括号内条件成立时返回1，否则返回0；[Image Omitted] 分别为页面 [Image Omitted] 的组件角色集合；若相似度达到预设阈值，则判定为重复页面，跳过后续处理；否则将页面签名加入已访问集合 [Image Omitted]；

枚举当前页面的可操作动作集合 [Image Omitted]，逐一执行动作 [Image Omitted]，观察生成的新页 [Image Omitted] 面，通过启发式规则识别弹窗、权限提示等临时页面，若 [Image Omitted] 的可操作组件数较少、存在取消、允许等特征文本且执行返回动作后可回到原页面 [Image Omitted]，则判定为临时页面，不记录跳转关系；否则记录跳转关系([Image Omitted])，并将 [Image Omitted] 入队 [Image Omitted]，执行系统返回动作回到页面 [Image Omitted]；

基于上述探索结果构建初始知识图谱 [Image Omitted] 为所有唯一页面对应的节点，以页面签名为唯一标识，[Image Omitted] 为控件树中可见且可交互的组件节点，每个组件节点满足：

[Image Omitted]

其中 [Image Omitted] 为组件显示文本，[Image Omitted] 为组件唯一标识，[Image Omitted] 为组件坐标范围，[Image Omitted] 分别为组件的交互属性，[Image Omitted] 为组件类型属性。

5.如权利要求4所述的基于语义知识图谱与LLM增强推理的移动软件自动化方法，其特征在于，语义增强与知识富集步骤包括：

引入轻量级大语言模型及场景化提示模板，对初始知识图谱 [Image Omitted] 中的节点与边进行语义补全；针对每个组件节点 [Image Omitted]，基于其原始属性 [Image Omitted]，通过下式推断功能语义角色：

[Image Omitted]

其中，[Image Omitted] 表示给定组件原始属性 [Image Omitted] 时，组件属于角色c的条件概率分布；c表示预定义的组件功能角色；[Image Omitted] 表示组件v的原始属性集合，包括 [Image Omitted] 等基础属性；[Image Omitted] 表示大语言模型，用于基于原始属性进行语义推理。将置信度最高的角色 [Image Omitted] 及置信值存入节点属性；

对组件的文本属性进行规范化处理，由 [Image Omitted] 生成语义等效的别名集合用于支撑后续模糊匹配；结合页面的结构特征与跳转历史，通过公式将页面划分为正常、临时、循环三类，[Image Omitted] 还将推断探索过程中未直接观察到的组件间逻辑关联及跨页面功能依赖，将这些语义属性与补全关系写入图数据库，最终形成增强知识图谱 [Image Omitted]。

6.如权利要求5所述的基于语义知识图谱与LLM增强推理的移动软件自动化方法，其特征在于，动态内容处理步骤包括：

遍历完整知识图谱 [Image Omitted] 中的节点，若超过预设值的多个节点具有相同的父节点、布局结构、角色类型，仅文本、图像属性存在差异，且所在区域支持滚动操作，则判定为该动态区域；为每个动态区域创建模板节点T，将模板节点存入 [Image Omitted]，删除动态区域中重复的实例节点，仅保留模板节点以控制图谱规模；在执行与监控步骤的自动执行过程中，通过系统级命令实时提取当前页面中的动态实例 [Image Omitted]，通过结构相似度匹配将实例 [Image Omitted] 与对应的模板节点T绑定；当用户指令涉及动态目标时，通过公式计算实例与指令的匹配得分：

[Image Omitted]

其中 [Image Omitted] 表示动态实例 [Image Omitted] 与用户指令查询目标 [Image Omitted] 的匹配得分；[Image Omitted] 表示文本相似度函数，用于计算实例文本 [Image Omitted] 与查询目标 [Image Omitted] 的文本重叠程度；[Image Omitted] 表示动态实例 [Image Omitted] 的动

态文本属性；[Image Omitted]表示语义特征相似度函数，用于计算实例语义特征 [Image Omitted] 与查询目标 [Image Omitted] 的语义相关性；[Image Omitted]表示动态实例 [Image Omitted] 的语义特征集合；[Image Omitted]表示实例 [Image Omitted] 在动态区域中的位置权重，按显示顺序赋予递减权重，筛选得分最高的实例作为交互目标。

7.如权利要求6所述的基于语义知识图谱与LLM增强推理的移动软件自动化方法，其特征在于，指令解析与图约束推理步骤包括：

基于完整知识图谱 [Image Omitted] 检索与该交互目标，进行多维度语义接地，通过公式计算 候选节点与指令的匹配得分：[Image Omitted]，其中，[Image Omitted]表示从知识图谱中检索到的候选节点；[Image Omitted]表示标准化后的用户指令查询目标；[Image Omitted]分别为文本相似度、语义嵌入相似度、角色兼容性、上下文相关性的权重系数；选取得分最高的节点作为接地目标节点 [Image Omitted]；

将任务完成过程建模为 [Image Omitted] 上的路径搜索问题，以当前UI状态对应的节点 [Image Omitted] 为起点，以接地目标节点 [Image Omitted] 为终点，采用最短路径搜索算法，搜索由合法动作边构成的最优路径，将路径转化为包含点击、输入、滑动的操作的指令操作序列。

8.如权利要求7所述的基于语义知识图谱与LLM增强推理的移动软件自动化方法，其特征在于，执行与监控步骤包括：

将指令操作序列转化为系统级输入事件，执行过程中，每完成一步操作后，通过系统级命令提取当前页面的控件树，计算页面签名 [Image Omitted]，与预期页面签名 [Image Omitted] 进行比对，若满足 [Image Omitted]， $\text{sim}()$ 表示前文定义的页面签名相似度函数， θ 表示状态匹配阈值；则判定当前操作执行成功，继续执行下一动作，否则触发异常处理机制；

异常处理机制包括：识别异常类型，若异常类型为临时弹窗，则执行关闭操作后继续原序列；若异常类型为跳转失败或动态实例变化，则重复执行当前动作，重试次数不超过预设次数，重试失败则基于 [Image Omitted] 重新规划路径。若异常类型为超时异常，则终止执行；

当前页面状态与目标节点对应的页面状态满足匹配阈值，且符合用户指令的核心意图时，判定任务执行成功。

9.一种基于语义知识图谱与LLM增强推理的移动软件自动化装置，其特征在于，包括：

初始知识图谱构建模块，提取移动设备运行移动应用APP时的界面信息，根据界面信息中每个访问的页面的内容和功能，探索得到每个访问页面的唯一页面签名，以构建初始知识图谱 [Image Omitted]，其中节点集V包含页面节点与其交互组件节点，边集E包含反映父子关系的结构边与记录操作跳转的动作边；

语义增强与知识富集模块，根据轻量级大语言模型LLM及场景化提示模板，对初始知识图谱 [Image Omitted] 进行深度语义补全，利用LLM，根据组件的属性推断节点的功能语义角色，并对节点的文本进行归一化处理，生成一组等效语义，以支撑的模糊匹配；结合初始知识图谱 [Image Omitted] 的结构，由LLM对界面类型进行分类，得到其属于正常、临时或循环类别；结合界面类型、等效语义、功能语义角色，利用LLM推断在探索中未直接观察到的组件间逻辑关联及跨页面功能依赖，从而补全该初始知识图谱 [Image Omitted] 的知识关系，得到语义增强的完整知识图谱 [Image Omitted]；

动态内容处理模块，在完整知识图谱 [Image Omitted] 中引入模板实例分离的表示机制，识别完整知识图谱 [Image Omitted] 的动态区域，作为结构模板节点 [Image Omitted]，记录其固定的布局模式、组件角色和语义标签；具体的动态项作为运行时实例 [Image Omitted]，包含实时变化的文本、图像特征，并通过动态绑定机制与相应模板节点 [Image Omitted] 关联；

指令解析与图约束推理模块，接收用户输入的自然语言指令 [Image Omitted]，并进行解析，进行多维度语义接地，基于完整知识图谱 [Image Omitted]，综合文本相似度、语义嵌入相关性、组件角色兼容性及其界面上下文等多个维度，计算自然语言指令 [Image Omitted] 与候选UI元素的匹配得分 [Image Omitted]，选择得分最高的候选UI元素作为目标组件；进行图基动作规划，将任务完成过程建模为在 [Image Omitted] 上的路径搜索问题，以当前状态节点为起点、接地目标节点为终点，搜索由合法动作边构成的最短路径，并生成指令操作序列；

执行与监控模块，将该指令操作序列的抽象动作序列转换为包括多步操作的系统级输入事件，以在该移动设备上自动执行，执行过程中，监控界面信息变化，在每一步操作后重新获取当前控件树、计算页面签名，并与预期状态进行比对；若检测到非预期弹窗、跳转失败或超时等情况，模块自动触发异常处理机制，采取关闭弹窗、操作重试或路径重新规划，以完成该自然语言指令 [Image Omitted]。

10.一种电子设备，其特征在于，包括权利要求9所述的一种基于语义知识图谱与LLM增强推理的移动软件自动化装置，该电子设备或连接有信息显示设备，该信息显示设备用于以用户设置的显示参数、属性或通过人工智能模型，显示该自然语言指令的执行结果。

11.一种计算机可读存储介质，其上存储有计算机程序，该计算机程序被处理器执行时实现权利要求1-8中任一所述基于语义知识图谱与LLM增强推理的移动软件自动化方法的步骤。

12.一种计算机程序产品，包括计算机程序，其特征在于，该计算机程序被处理器执行时实现权利要求1-8中任一所述基于语义知识图谱与LLM增强推理的移动软件自动化方法的步骤。

Language: (English)

1. A mobile software automation method based on semantic knowledge graph and LLM enhanced reasoning is characterized by comprising the following steps:

An initial knowledge graph construction step, namely extracting interface information when mobile equipment runs mobile Application (APP), and exploring and obtaining a unique page signature of each access page according to the content and the function of each access page in the interface information to construct an initial knowledge graph [Image Omitted] The node set V comprises page nodes and interaction component nodes thereof, and the edge set E comprises structural edges reflecting father-son relations and action edges for recording operation jumps;

Semantic enhancement and knowledge enrichment, namely, according to a lightweight large language model LLM and a scene prompt template, performing initial knowledge graph [Image Omitted] Performing deep semantic completion, deducing functional semantic roles of the nodes according to the attributes of the components by using LLM, normalizing the text of the nodes to generate a group of equivalent semantics for supporting fuzzy matching, combining the initial knowledge graph [Image Omitted] Classifying interface types by LLM to obtain the interface types belonging to normal, temporary or cyclic categories, combining interface types, equivalent semantics and functional semantics roles, and utilizing LLM to infer logical association among components and cross-page function dependence which are not directly observed in exploration so as to complement the initial knowledge graph [Image Omitted] Obtaining a semantically enhanced complete knowledge graph [Image Omitted] ;

Dynamic content processing step, in the complete knowledge graph [Image Omitted] Introducing a representation mechanism of template instance separation, and identifying a complete knowledge graph [Image Omitted] As a structural template node [Image Omitted] Recording its fixed layout mode, component role and semantic label, and using specific dynamic item as run-time example [Image Omitted] Contains the text and image characteristics which change in real time and is connected with the corresponding template nodes through a dynamic binding mechanism [Image Omitted] Associating;

instruction analysis and graph constraint reasoning step, receiving natural language instruction input by user [Image Omitted] Analyzing, performing multidimensional semantic grounding, and based on a complete knowledge graph [Image Omitted] Calculating natural language instructions by integrating multiple dimensions such as text similarity, semantic embedding correlation, component role compatibility and interface context [Image Omitted] Matching score with candidate UI elements [Image Omitted] Selecting candidate UI elements with highest scores as target components, performing graph-based action planning, and modeling a task completion process as in [Image Omitted] The path searching problem searches the shortest path formed by legal action edges by taking the current state node as a starting point and the grounding target node as an end point, and generates an instruction operation sequence;

Executing and monitoring, namely converting the abstract action sequence of the instruction operation sequence into a system-level input event comprising multi-step operation so as to automatically execute the abstract action sequence on the mobile device, and monitoring the change of interface information in the executing process, re-acquiring a current control tree, calculating page signatures and comparing the current control tree with an expected state after each step of operation [Image Omitted] 。

2. The mobile software automation method based on semantic knowledge graph and LLM enhanced reasoning of claim 1, wherein the performing and monitoring steps comprise:

a knowledge graph increment maintenance step, in which newly discovered interface states, components and interaction relations are continuously fed back and updated to complete knowledge graphs in the automatic execution process [Image Omitted] The online learning and incremental expansion of the atlas are realized, so that the system has the capability of continuous evolution.

3. The mobile software automation method based on semantic knowledge graph and LLM enhanced reasoning of claim 1, wherein the initial knowledge graph construction step comprises the steps of exploring the content and function of each accessed page in the interface

information to ensure the initial knowledge graph [Image Omitted] Pruning the interface information, including integrating the contract-level information through text merging, transmitting dynamic state through attribute inheritance mechanism, and eliminating invalid nodes with empty text, messy codes and abnormal coordinates.

4. The mobile software automation method based on semantic knowledge graph and LLM enhanced reasoning of claim 1, wherein the initial knowledge graph construction step comprises:

Initial knowledge graph [Image Omitted] Center node set [Image Omitted] Edge set [Image Omitted] , [Image Omitted] Is a structural side of the structure, and is provided with a plurality of grooves, [Image Omitted] Is an action edge, and when the APP program runs, an application initial starting page is started [Image Omitted] Inclusion exploration queue [Image Omitted] I.e. [Image Omitted] Initializing a set of accessed page signatures [Image Omitted] The action edge types comprise clicking, text input, sliding and returning operations;

Queue pair [Image Omitted] Each page of (3) [Image Omitted] The control tree is extracted by a system level command [Image Omitted] Calculating the unique page signature based on the layout structure, the text content and the component type as:

[Image Omitted]

wherein, the [Image Omitted] Representing pages [Image Omitted] The unique structure signature of the (a) is used for page duplication removal and identity recognition; [Image Omitted] representing pages [Image Omitted] The text token set of (2) is generated by normalizing visible component text; [Image Omitted] representing pages [Image Omitted] By mapping control trees to layout hash values of (a) [Image Omitted] Performing hash calculation to obtain the data for capturing the parent-child hierarchical relationship and ordering characteristics of the components; [Image Omitted] representing pages [Image Omitted] The component role sets are extracted from the component attributes and are determined after LLM judgment and used for representing the component function types;

Calculating the similarity of the page signatures through a weighted combination formula, and judging whether the page has been accessed or not:

[Image Omitted]

wherein, the [Image Omitted] Representing pages [Image Omitted] Is signed by (a) [Image Omitted] And page [Image Omitted] Is signed by (a) [Image Omitted] The similarity between the two is in the value range of 0, 1; [Image Omitted] the weight coefficients of the text similarity, the layout consistency and the role similarity respectively satisfy the following conditions [Image Omitted] ; [Image Omitted] Representing a Jacquard similarity function for calculating the degree of overlap of the two sets; [Image Omitted] Respectively are pages [Image Omitted] A text token set of (a); [Image Omitted] Respectively are pages [Image Omitted] $1[-j]$ is an indication function, returns 1 when the condition in brackets is satisfied, otherwise returns 0; [Image Omitted] Respectively are pages [Image Omitted] If the similarity reaches a preset threshold, determining to repeat the page, skipping the subsequent processing, otherwise, adding the page signature into the accessed set [Image Omitted] ;

Enumerating an operational action set for a current page [Image Omitted] Executing actions one by one [Image Omitted] Observing the new page generated [Image Omitted] Face, identify the temporary pages such as popup window, permission prompt, etc. through heuristic rules, if [Image Omitted] Fewer number of operable components, cancellation, permission of feature text, etc., and return to the original page after performing the return action [Image Omitted] Judging that the page is a temporary page, not recording the jump relation, otherwise recording the jump relation [Image Omitted]) And will [Image Omitted] Enqueue [Image Omitted] Executing a system return action back to the page [Image Omitted] ;

Constructing an initial knowledge graph based on the exploration result [Image Omitted] For all the nodes corresponding to the unique pages, the page signature is taken as the unique identification, [Image Omitted] For visible and interactable component nodes in the control tree, each component node satisfies:

[Image Omitted]

Wherein the method comprises the steps of [Image Omitted] The text is displayed for the component and, [Image Omitted] Is a unique identification for the component and, [Image Omitted] For a range of coordinates of the component, [Image Omitted] The interaction attributes of the components are respectively, [Image Omitted] Is a component type attribute.

5. The automated mobile software method based on semantic knowledge graph and LLM enhanced reasoning of claim 4, wherein the semantic enhancement and knowledge enrichment steps comprise:

a lightweight large language model and a scene prompt template are introduced, and an initial knowledge graph is mapped [Image Omitted] Semantic completion of the nodes and edges in the model, and aiming at each component node [Image Omitted] Based on its original properties [Image Omitted] Deducing functional semantic roles by:

[Image Omitted]

wherein, the [Image Omitted] Representing original properties of a given component [Image Omitted] When the component belongs to the conditional probability distribution of the role c , c represents a predefined component function role; [Image Omitted] Representing the original set of properties of component v , including [Image Omitted] Basic attributes are equal; [Image Omitted] a large language model is represented for semantic reasoning based on the original attributes. Roles with highest confidence [Image Omitted] Storing the confidence value into the node attribute;

normalizing text attribute of component, comprising [Image Omitted] Generating semantically equivalent alias sets for supporting subsequent fuzzy matching, combining structural features of the pages and jump histories, dividing the pages into three types of normal, temporary and cyclic through formulas, [Image Omitted] The logical association and the cross-page function dependence among components which are not directly observed in the inferred exploration process are written into the graph database, and finally the enhanced knowledge graph is formed [Image Omitted] .

6. The mobile software automated method based on semantic knowledge graph and LLM enhanced reasoning of claim 5, wherein the dynamic content processing step comprises:

traversing the complete knowledge graph [Image Omitted] If the nodes exceeding the preset value have the same father node, layout structure and role type, only the text and image attributes have difference, and the area supports the scrolling operation, judging the dynamic area [Image Omitted] In the automatic execution process of the execution and monitoring steps, the dynamic instance in the current page is extracted in real time through a system level command [Image Omitted] Instances are matched by structural similarity [Image Omitted] Binding with the corresponding template node T , and calculating the matching score of the instance and the instruction through a formula when the user instruction relates to the dynamic target:

[Image Omitted]

Wherein the method comprises the steps of [Image Omitted] Representing dynamic instances [Image Omitted] Query targets with user instructions [Image Omitted] Matching scores of (2); [Image Omitted] representing text similarity functions for computing instance text [Image Omitted] With query targets [Image Omitted] Is a text overlap degree of (2); [Image Omitted] representing dynamic instances [Image Omitted] Dynamic text properties of (a); [Image Omitted] representing semantic feature similarity functions for computing instance semantic features [Image Omitted] With query targets [Image Omitted] Semantic relatedness of (c); [Image Omitted] representing dynamic instances [Image Omitted] Is a semantic feature set of (1); [Image Omitted] Representing examples [Image Omitted] And (3) giving a descending weight to the position weight in the dynamic area according to the display sequence, and screening the example with the highest score as an interaction target.

7. The mobile software automated method based on semantic knowledge graph and LLM enhanced reasoning of claim 6, wherein the instruction parsing and graph constraint reasoning step comprises:

based on complete knowledge graph [Image Omitted] Searching the interaction target, carrying out multidimensional semantic grounding, and calculating a matching score of the candidate node and the instruction through a formula: [Image Omitted] , wherein, [Image Omitted] Representing candidate nodes retrieved from the knowledge graph; [Image Omitted] Representing a standardized user instruction query target; [Image Omitted] the weight coefficients of text similarity, semantic embedding similarity, role compatibility and context correlation are respectively; selecting the node with the highest score as a grounding target node [Image Omitted] ;

Modeling a task completion process as [Image Omitted] The path searching problem on the node corresponding to the current UI state [Image Omitted] Starting from a grounded target node [Image Omitted] And searching an optimal path formed by legal action edges by adopting a shortest path searching algorithm as an end point, and converting the path into an instruction operation sequence comprising clicking, inputting and sliding operations.

8. The mobile software automation method based on semantic knowledge graph and LLM enhanced reasoning of claim 7, wherein the performing and monitoring steps comprise:

converting instruction operation sequences into system-level input events, extracting a control tree of a current page through a system-level command after each operation is completed in the execution process, and calculating page signatures [Image Omitted] Signature with expected page [Image Omitted] Comparing, if it meets [Image Omitted] The $\text{sim}()$ represents the page signature similarity function defined before, and the θ represents the state matching threshold value, so that the current operation is judged to be successfully executed, the next action is continuously executed, and otherwise, an exception handling mechanism is triggered;

the exception handling mechanism comprises identifying exception type, if the exception type is a temporary popup window, executing closing operation and then continuing the original sequence, if the exception type is jump failure or dynamic instance change, repeatedly executing current action, wherein the retry frequency is not more than preset frequency, and the retry failure is based on the result of the follow-up operation [Image Omitted] The path is re-planned. If the exception type is overtime exception, stopping execution;

and when the current page state meets a matching threshold value with the page state corresponding to the target node and accords with the core intention of the user instruction, judging that the task is successfully executed.

9. A mobile software automation device based on semantic knowledge graph and LLM enhanced reasoning, comprising:

The initial knowledge graph construction module extracts interface information when the mobile equipment runs the mobile application APP, and searches and obtains a unique page signature of each access page according to the content and the function of each access page in the interface information so as to construct an initial knowledge graph [Image Omitted] The node set V comprises page nodes and interaction component nodes thereof, and the edge set E comprises structural edges reflecting father-son relations and action edges for recording operation jumps;

the semantic enhancement and knowledge enrichment module is used for carrying out initial knowledge graph according to a lightweight large language model LLM and a scene prompt template [Image Omitted] Performing deep semantic completion, deducing functional semantic roles of the nodes according to the attributes of the components by using LLM, normalizing the text of the nodes to generate a group of equivalent semantics for supporting fuzzy matching, combining the initial knowledge graph [Image Omitted] Classifying interface types by LLM to obtain the interface types belonging to normal, temporary or cyclic categories, combining interface types, equivalent semantics and functional semantics roles, and utilizing LLM to infer logical association among components and cross-page function dependence which are not directly observed in exploration so as to complement the initial knowledge graph [Image Omitted] Obtaining a semantically enhanced complete knowledge graph [Image Omitted] ;

Dynamic content processing module, in complete knowledge graph [Image Omitted] Introducing a representation mechanism of template instance separation, and identifying a complete knowledge graph [Image Omitted] As a structural template node [Image Omitted] Recording its fixed layout mode, component role and semantic label, and using specific dynamic item as run-time example [Image Omitted] Contains the text and image characteristics which change in real time and is connected with the corresponding template nodes through a dynamic binding mechanism [Image Omitted] Associating;

the instruction analysis and graph constraint reasoning module receives natural language instructions input by a user [Image Omitted] Analyzing, performing multidimensional semantic grounding, and based on a complete knowledge graph [Image Omitted] Calculating natural language instructions by integrating multiple dimensions such as text similarity, semantic embedding correlation, component role compatibility and interface context [Image Omitted] Matching score with candidate UI elements [Image Omitted] Selecting candidate UI elements with highest scores as target components, performing graph-based action planning, and modeling a task completion process as in [Image Omitted] The path searching problem searches the shortest path formed by legal action edges by taking the current state node as a starting point and the grounding target node as an end point, and generates an instruction operation sequence;

the execution and monitoring module converts the abstract action sequence of the instruction operation sequence into a system-level input event comprising multi-step operation so as to be automatically executed on the mobile equipment, monitors the change of interface information in the execution process, re-acquires the current control tree after each step of operation, calculates page signature, compares the current control tree with the expected state, and automatically triggers an exception handling mechanism by the module if unexpected popup, jump failure or overtime and the like are detected, and adopts closing popup, operation retry or path re-planning to complete the natural language instruction [Image Omitted] .

10. An electronic device, characterized by comprising the mobile software automation device based on semantic knowledge graph and LLM enhanced reasoning as claimed in claim 9, wherein the electronic device is connected with an information display device, and the information display device is used for displaying the execution result of the natural language instruction by using display parameters, attributes or through an artificial intelligence model set by a user.

11. A computer readable storage medium having stored thereon a computer program which, when executed by a processor, implements the steps of the mobile software automation method based on semantic knowledge graph and LLM enhanced reasoning of any of claims 1-8.

12. A computer program product comprising a computer program, characterized in that the computer program, when executed by a processor, implements the steps of the mobile software automation method based on semantic knowledge graph and LLM enhanced reasoning as claimed in any one of claims 1-8.
